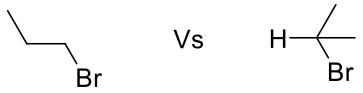
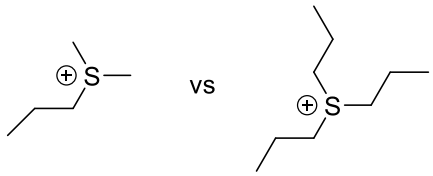
Questions Set-2_CH2105

1. Predict which reaction in each of the following pairs is expected to have a higher ratio of elimination to substitution. Cite the possible reason.

i) 

(Reaction condition: NaOEt/EtOH)

ii) 

(Reaction condition: NaOEt/EtOH)

iii)

(Reaction condition: NaOEt/EtOH)

Solutions: i) Primary halides are more prone towards substitution (steric factor) whereas secondary and tertiary halides prefer to undergo elimination mainly because of the steric bulk and more importantly due to the formation of more substituted double bond (Zaitsev rule). So, 2>1

ii) Due to the more number of beta hydrogen, more probability of elimination is there in 2 than 1.

iii) In 1, there is not a single beta hydrogen anti to the halogen (important criteria for elimination), whereas in 2, there is one of that type. So in 1 there is no elimination, but in 2 elimination will happen.

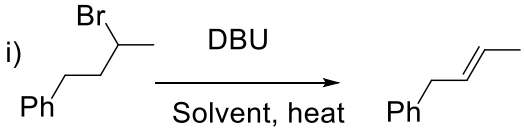
2. Write the rate equation for E1, E2, and E1CB mechanisms.

Solutions: E1→ rate=k[R-X]

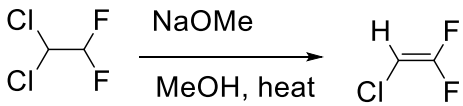
E2→ rate=k[R-X][base]

E1cB→ rate=k_1k_2[R-X][base]/k_{-1}[CB]+k_2 **CB = conjugate base**

3. The following reactions proceed via the E2 and E1CB pathways. Which experiments will you perform to establish the correct mechanism?

i) 

(It follows E2 path for major product)

ii) 

(It follows E1CB path for major product)

Solutions:

Mechanism of E1cB

The E2 mechanism is ruled out when the kinetic isotope effect is about 1.0, which shows that the C-H bond is not broken in the rate-determining phase.

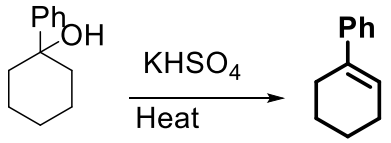
The E2 mechanism

The coordinated mechanism of E2 reactions can be verified by deuterium isotope experiments. The deuterium-labeled molecule will react more slowly if a C-H bond is

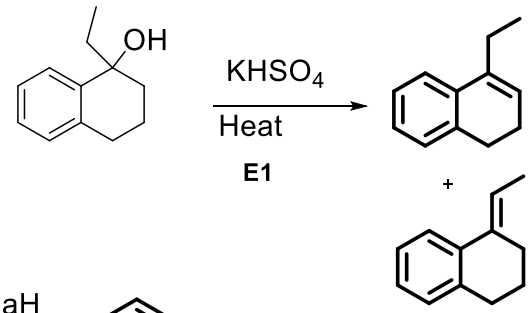
broken in the rate-determining phase because the C-D bond has a higher energy than the C-H bond.

4. Write the major product of the following reactions.

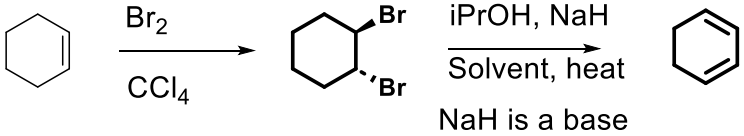
i)



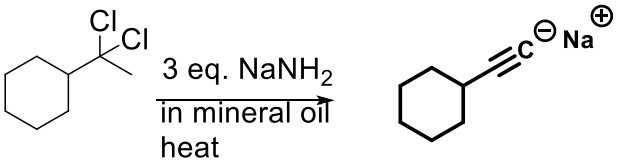
ii)



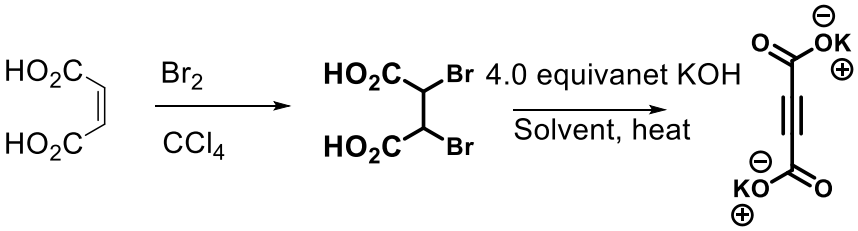
iii)



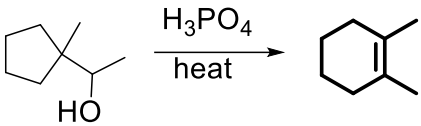
iv)



v)

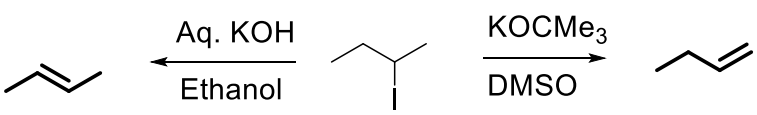


vi)



(rearrangement, ring expansion via E1 mehanism, 3 degree carbocation is more stable than 2 degree carbocation)

vii)



(less hindered base promotes Zeitsev product) **(more hindered base promotes Hofmann product)**